PyTorch 1: Building Your First Neural Network

Thursday, 1 October 2026 5:11 PM

PyTorch → Is a Deep learning Framework, developed by Meta.

(FAIR)

12 Columns.

1 row → 1 person.

Data description

U

$$2x_1 + 2x_2 + 2x_3$$
$$2 \cdot 1 + 2 \cdot 2 + 2 \cdot 3$$
$$2 + 4 + 6.$$

y is the output

x is the input.
↳ is the variable we are asking the gradient about.

x.requires_grad = True

Tell PyTorch to track the gradient for x.

x.grad

Gradient of y wrt x.

y.backward

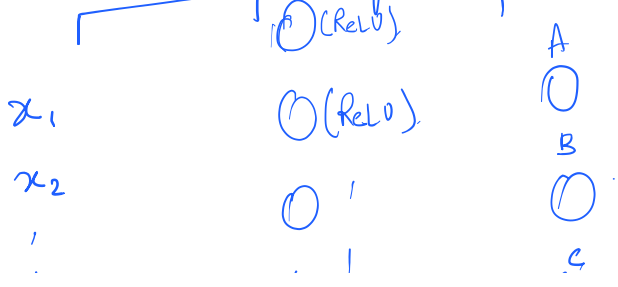
Start the backprop from y.

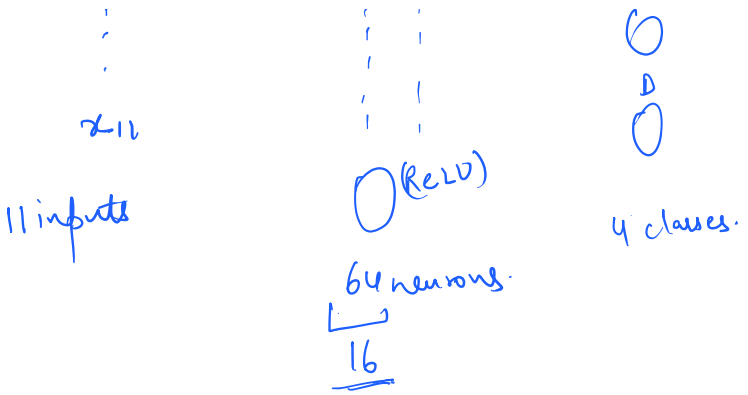
* nn.Layers.

Use to created a dense layer.

in_features = 11 , out_features = 64

$W_{11 \times 64}$ $b = 64$. $W_{64 \times 4}$ $b = 4$





* Weight Initialization.

1) Kaiming uniform (He)

ReLU or Leaky ReLU

Kaiming uniform (the default for `nn.Linear`):

$$w \sim \text{Uniform}\left[-\frac{1}{\sqrt{fan_in}}, \frac{1}{\sqrt{fan_in}}\right]$$

$\sqrt{10}$ $\sqrt{10}$

2) Glorot (Xavier) uniform

Sigmoid / Tanh Activation

Glorot (Xavier) uniform (a widely-used alternative, common across many libraries and papers):

$$w \sim \text{Uniform}\left[-\frac{\sqrt{6}}{\sqrt{fan_in + fan_out}}, \frac{\sqrt{6}}{\sqrt{fan_in + fan_out}}\right]$$

$\sqrt{10+5}$ $\sqrt{10+5}$.

fan_in = # of inputs coming into the layer.

fan_out = # of outputs going out of this layer.

nn.Linear(10, 5)

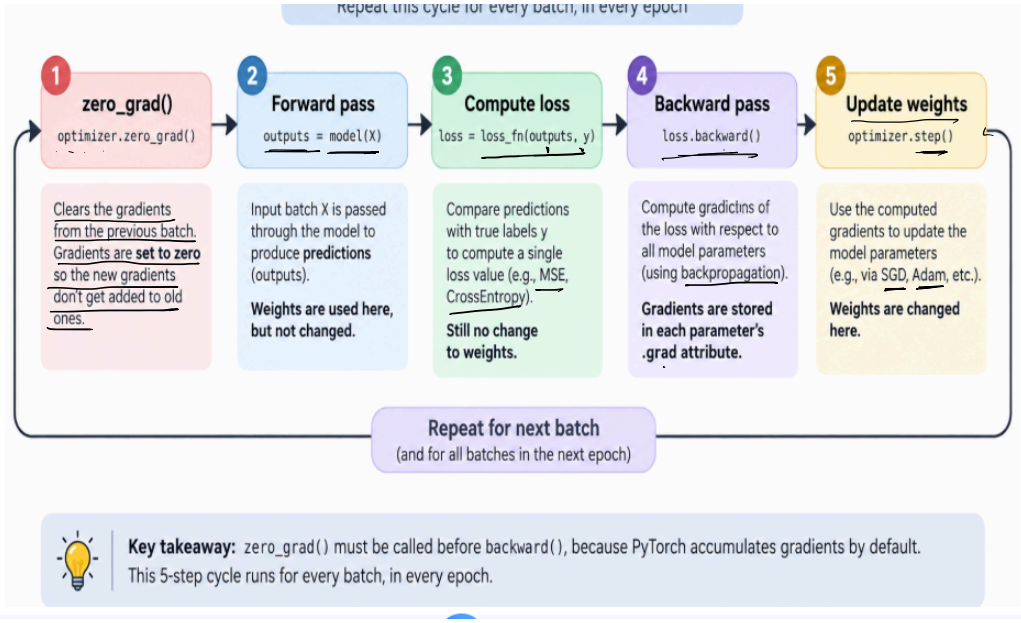
↑ fan_in ↑ fan_out.

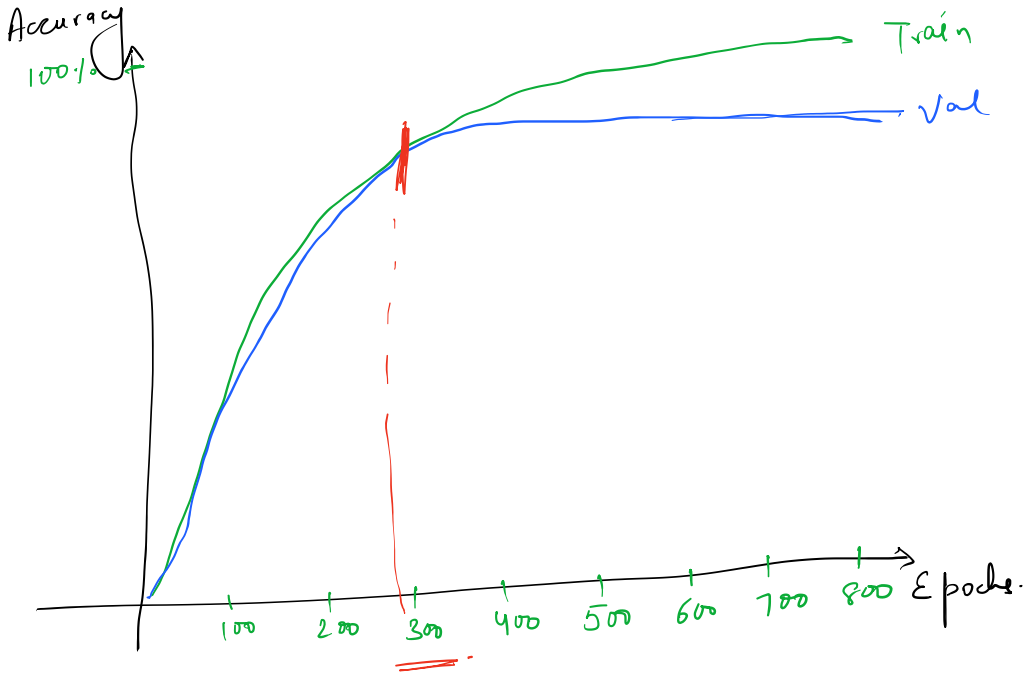
epoch 1. $W^{new} = W^{old} - \eta * \text{grad}_1$

epoch 2 $W^{new} = W^{old} - \eta * \text{grad}_2$.

The Training Loop (for each batch)

Repeat this cycle for every batch, in every epoch





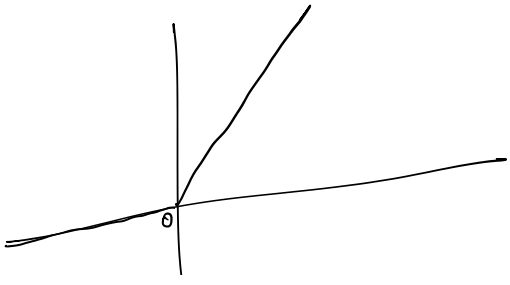
$Z = 1000.$

$\text{ReLU} = \max(0, Z)$

No trainable parameter.

pre activated output.

$(W^T X + b) = Z.$



1) Time Series Forecasting.

Model → plant.



15 years data.

→ F.
N months.

Prophet.

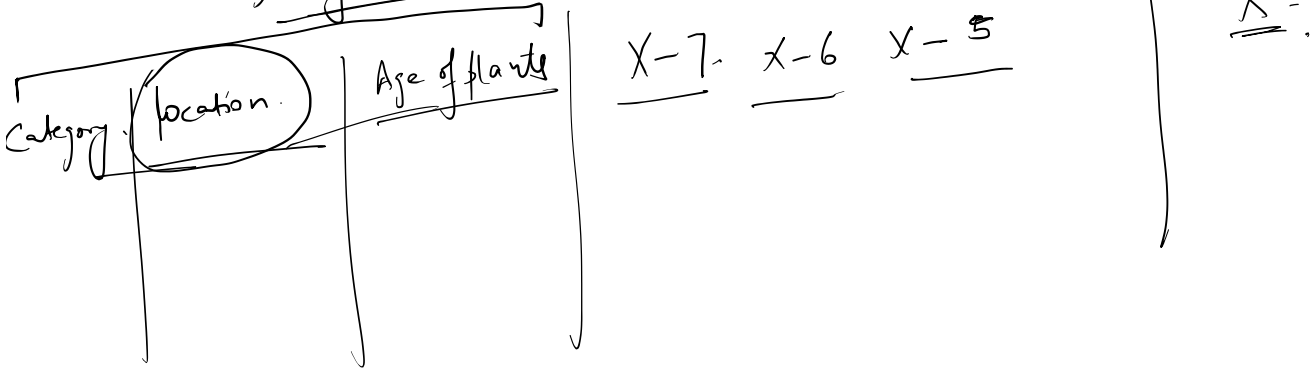
VCM.

ARIMA / ARIMAX.

trend
seasonality.

M
Y . 5

2) Regression



3) LSTMs-